



Bibliography tracking

The following table provides an overview of the references used in the thesis, indicating whether each work has been read and whether it is currently cited in the written material.

Table 1: Tracking of GEO datasets, manifests, and probe-filtering resources

First author / Source	Dataset / Resource title	Read	Used	Ref.
GEO datasets				
NCBI	GSE69914: DNA methylation profiles in breast tissue samples	Yes	Yes	[1]
NCBI	GSE225845: High neighborhood deprivation impacts DNA methylation and gene expression in cancer-related genes	Yes	Yes	[2]
NCBI	GSE287331: DNA methylation patterns in breast cancer, benign tissue, and healthy controls	Yes	Yes	[3]
Dennis	DNA methylation patterns in breast cancer, paired benign tissue from ipsilateral and contralateral breast, and healthy controls	No	Yes	[4]
Manifests & annotation resources				
Illumina	Infinium MethylationEPIC v1.0 B5 Manifest File	Yes	Yes	[5]
Robinson	IlluminaHumanMethylation450kanno.ilmn12.hg19: annotation package for 450k arrays (Bioconductor)	Yes	Yes	[6]
Probe filtering & array-specific issues				
Naeem	Reducing the risk of false discovery enabling identification of biologically significant genome-wide methylation status using the HM450 array	Yes	Yes	[7]
Chen	Discovery of cross-reactive probes and polymorphic CpGs in the Illumina HumanMethylation450 microarray	Yes	Yes	[8]
Pidsley	Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling	No	Yes	[9]
Zhou	Comprehensive characterization, annotation and innovative use of Infinium DNA methylation BeadChip probes	Yes	Yes	[10]
McCartney	Identification of polymorphic and off-target probe binding sites on the Illumina Infinium MethylationEPIC BeadChip	Yes	Yes	[11]

Table 2: Tracking of consulted and cited references for Medical / biological background

First author	Full paper title	Read	Cited	Ref.
Medical / biological background				
NIH	Understanding Cancer	Yes	No	[12]
NCI	BRCA Gene Changes: Cancer Risk and Genetic Testing Fact Sheet	Yes	Yes	[13]
Azizi	Introduction to Cancer Biology	Yes	No	[14]
Menon	Breast Cancer (StatPearls)	Yes	No	[15]
Hanahan	The Hallmarks of Cancer	Yes	No	[16]
Ehrlich	DNA Methylation in Cancer: Too Much, but Also Too Little	Yes	Yes	[17]
Bird	DNA Methylation Patterns and Epigenetic Memory	Yes	Yes	[18]
Russo	Epigenetic Mechanisms of Gene Regulation	Yes	Yes	[19]
Pina-Sanchez	Cancer Biology, Epidemiology, and Treatment in the 21st Century: Current Status and Future Challenges From a Biomedical Perspective	No	No	[20]
Wilcox	The CpG Landscape of Protein Coding DNA in Vertebrates	Yes	No	[21]
Ji	A comprehensive methylome map of lineage commitment from Hematopoietic Progenitors	No	No	[22]
Jenkins	Neighborhood Deprivation and DNA Methylation and Expression of Cancer Genes in Breast Tumors	No	No	[23]

Table 3: Tracking of consulted and cited references for Data visualization and exploration

First author	Full paper title	Read	Cited	Ref.
Data visualization and exploration				
Maksimovic	SWAN: Subset-quantile Within Array Normalization for Illumina HumanMethylation450 BeadChips	Yes	Yes	[24]
Benelli	Charting Differentially Methylated Regions in Cancer with Rocker-meth	Yes	No	[25]
Seebboth	DNA Methylation Outlier Burden, Health, and Ageing in Generation Scotland and the Lothian Birth Cohorts	Yes	No	[26]
Zhao	A Urine-based DNA Methylation Assay, ProCURE, to Identify Clinically Significant Prostate Cancer	Yes	No	[27]
Chen	Detecting Early-Warning Signals for Sudden Deterioration of Complex Diseases by Dynamical Network Biomarkers	No	Yes	[28]
West	On Dynamic Network Entropy in Cancer	No	Yes	[29]
Liu	Identifying the Critical States and Dynamic Network Biomarkers of Cancers Based on Network Entropy	No	Yes	[30]
Islam	Integration of DNA Methylation Patterns and Genetic Variation in Human Pediatric Tissues Helps Inform EWAS Design and Interpretation	No	Yes	[31]
Mansell	Guidance for DNA Methylation Studies: Statistical Insights from the Illumina EPIC Array	No	Yes	[32]
Chen	mLiftOver: Mapping DNA Methylation Data Across Array Platforms and Genome Builds	No	Yes	[33]
Yousefi	Considerations for normalization of DNA methylation data by Illumina 450K BeadChip assay in population studies	No	Yes	[34]
Liu	An evaluation of processing methods for Illumina HumanMethylation450 BeadChip platform	No	Yes	[35]
Welsh	A systematic evaluation of normalization methods and probe replicability using Infinium EPIC 850K methylation data	No	Yes	[36]
Pidsley	A data-driven approach to preprocessing Illumina 450K methylation array data	No	Yes	[37]
Maksimovic	A cross-package Bioconductor workflow for analysing methylation array data	No	Yes	[38]
Zhang	Critical evaluation of the reliability of DNA methylation probes on the Illumina MethylationEPIC v1.0 BeadChip microarrays	No	No	[39]
Liu	An evaluation of processing methods for HumanMethylation450 BeadChip data	No	No	[35]

Table 4: Tracking of consulted and cited references for GSE69914 Applications

First author	Full paper title	Read	Cited	Ref.
GSE69914 Applications				
Chen	Discovery of cross-reactive probes and polymorphic CpGs in the Illumina HumanMethylation450 microarray	Yes	No	[8]
Teschendorff	DNA methylation outliers in normal breast tissue identify field defects that are enriched in cancer	Yes	Yes	[40]
Gao	The integrative epigenomic-transcriptomic landscape of ER-positive breast cancer	Yes	No	[41]
Ding	Integrative analysis identifies potential DNA methylation biomarkers for pan-cancer diagnosis and prognosis	Yes	No	[42]
Gao	DNA methylation patterns in normal tissue correlate more strongly with breast cancer status than CNVs	Yes	No	[43]
Croes	Large-scale analysis of DFNA5 methylation reveals its potential as biomarker for breast cancer	Yes	No	[44]
Fan	Prediction of disease genes based on stage-specific gene regulatory networks in breast cancer	Yes	No	[45]
Gao	Genome-wide discovery of circulating cell-free DNA methylation signatures for TNBC diagnosis	Yes	No	[46]
Zhu	An improved epigenetic counter to track mitotic age in normal and precancerous tissues	Yes	No	[47]
Yang	An integrative pan-cancer-wide analysis of epigenetic enzymes reveals universal patterns of epigenomic deregulation in cancer	Yes	No	[48]
Bartlett	Inference of tissue relative proportions of breast epithelial subtypes (luminal progenitor, basal, luminal mature)	Yes	No	[49]
Teschendorff	EPISCORE: Cell-type deconvolution of bulk DNA methylomes using single-cell RNA-seq data	Yes	No	[50]

Table 5: Tracking of consulted and cited references for Data Preparation

First author	Full paper title	Read	Cited	Ref.
Data Preparation				
Teschendorff	A beta-mixture quantile normalization method for correcting probe design bias in Illumina Infinium 450k DNA methylation data	No	Yes	[51]
de Almeida	A multimodal conversational agent for DNA, RNA and protein tasks	No	Yes	[52]
Sokolov	Decoding depression: a comprehensive multi-cohort exploration of blood DNA methylation using machine learning and deep learning approaches	No	Yes	[53]

Table 6: Tracking of consulted and cited references for Data Pre-processing & quality control

First author	Full paper title	Read	Cited	Ref.
Data Pre-processing & quality control MANCANO PAPERSSSSS				
Du	Comparison of Beta-value and M-value methods for quantifying methylation levels by microarray analysis	Yes	Yes	[54]
Cazaly	Comparison of pre-processing methodologies for Illumina 450k methylation array data in familial analyses	Yes	No	[55]
Xu	The ENmix DNA methylation analysis pipeline for Illumina BeadChip and comparisons with seven other preprocessing pipelines	No	No	[56]
Bizet	Improving Infinium MethylationEPIC data processing: re-annotation of enhancers and long noncoding RNA genes and benchmarking of normalization methods	No	No	[57]
Morris	Analysis pipelines and packages for Infinium HumanMethylation450 BeadChip (450k) data	No	No	[58]

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